# Drone Technology & UAV Systems Internship- Task 1

UAV Fundamentals, Flight Control Basics & Simulation Setup

## Part 1 – Core Learning (Self Study Module)

### 1. What is UAV vs Drone:

A UAV (Unmanned Aerial Vehicle) is an aircraft without a human pilot onboard. A drone is a more general term used for UAVs that can be remotely controlled or autonomous.

2. Types of Drones:

- Multirotor: Most common drones with multiple rotors (e.g., quadcopters).
- Fixed-wing: Airplane-like drones, efficient for long-distance travel.
- Hybrid VTOL: Combine vertical takeoff and landing with forward flight.

| Specification | Description | | | | |
|---|---|---|---|---|---|
| | Multi-rotor | Helicopter | Fixed-wing | Blimps | Flying wing |
| | | | | | |
| Model | DJIS1000+ | AXH-E230 | Bat-3 | CB3000 | Pathfinder-Plus |
| Manufacturer | DJI technology | AVIX | MLB Co. | Beijing CSCA Co. | AeroVironment |
| Materials | Carbon fiber, High strength performance engineered plastics | Carbon fiber, aluminum alloy | Carbon fiber, engineered plastics | Kevlar fibers, fiber optic, electrical cores | Carbon fiber, Nomex, Kevlar, plastic sheeting, plastic foam |
| Cost | Low | Medium | Medium | High | very high |
| Power/Motors | Eight electric, 0.5 kw max each | One BLDC motors | Two-stroke engine | One oil engine | Eight (8) solar-electric, 1.5 kW max each |
| Gross weight/kg[a] | 6 | 15 | 56 | 300 | 318 |
| Payload capacity/kg[b] | 7 | 15 | 9 | 10 | 67.5 |
| Speed/m $s^{-1}$ | 12 | 23 | 33 | 15 | 14 |
| Endurance/h[c] | 0.25 | 0.8 | 6 | 12 | 15 |
| Altitude ceiling/m | 500[d] | 3,000 | 3,000 | 120 | 25,000 |

*[a] Total weight with a battery; [b] The payload including battery; [c] Endurance with maximum payload; [d] The maximum flight height in China (the flight control system was restricted by the national regulations to set the flight height lower than 500 m).*

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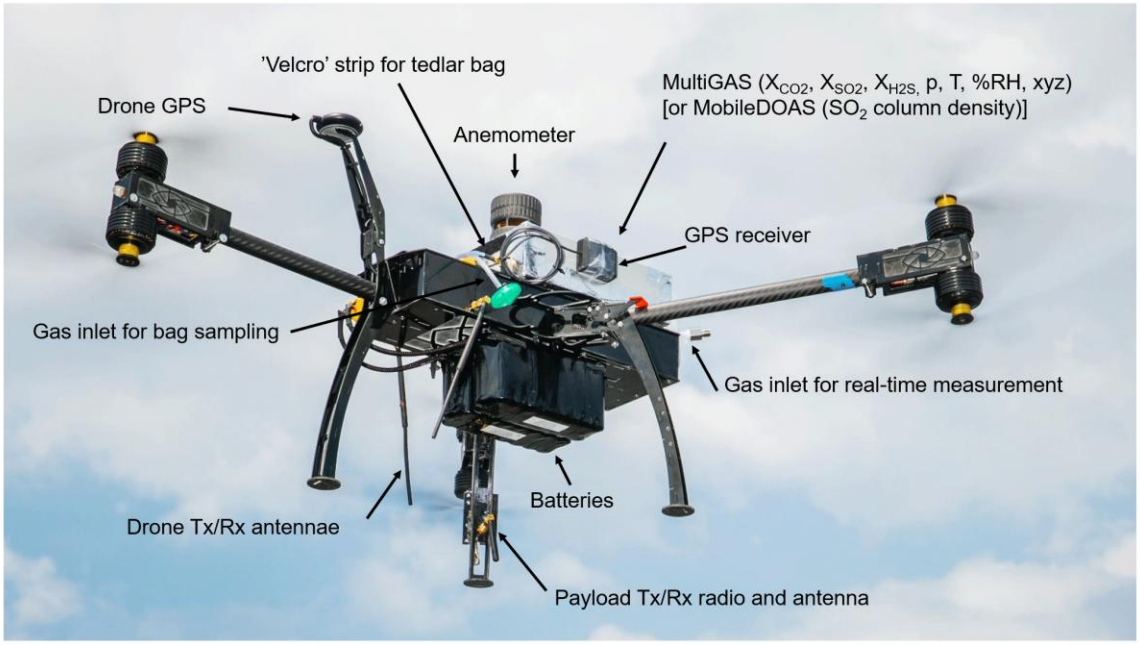
3. Basic Components:

- Flight Controller: Brain of the drone.
- ESC (Electronic Speed Controller): Controls motor speed.
- GPS Module: Provides location data.
- IMU: Measures acceleration and rotation.
- LiDAR/Camera: Used for sensing and imaging.

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4. Key Concepts:
- PID Control: Maintains stability.
- MAVLink: Communication protocol.
- Autopilot Firmware: Software controlling the drone.

**PID control stability** is influenced by the configuration of the controller:

PID Controller modes affect the overall system stability and Gain Margin. In general, if Proportional Mode is our reference, Integral (or Proportional + Integral) Control reduces relative stability of the closed loop system, while Proportional + Derivative Control increases relative stability of the closed loop system.

## MAVLINK

MAVLink (Micro Air Vehicle Link) is a lightweight, header-only message marshaling library used for communication between unmanned vehicles (drones) and Ground Control Stations (GCS). It supports bidirectional communication for telemetry and command, utilizing a publish-subscribe pattern to transfer data efficiently over low-bandwidth links.

## Autopilot Frimware

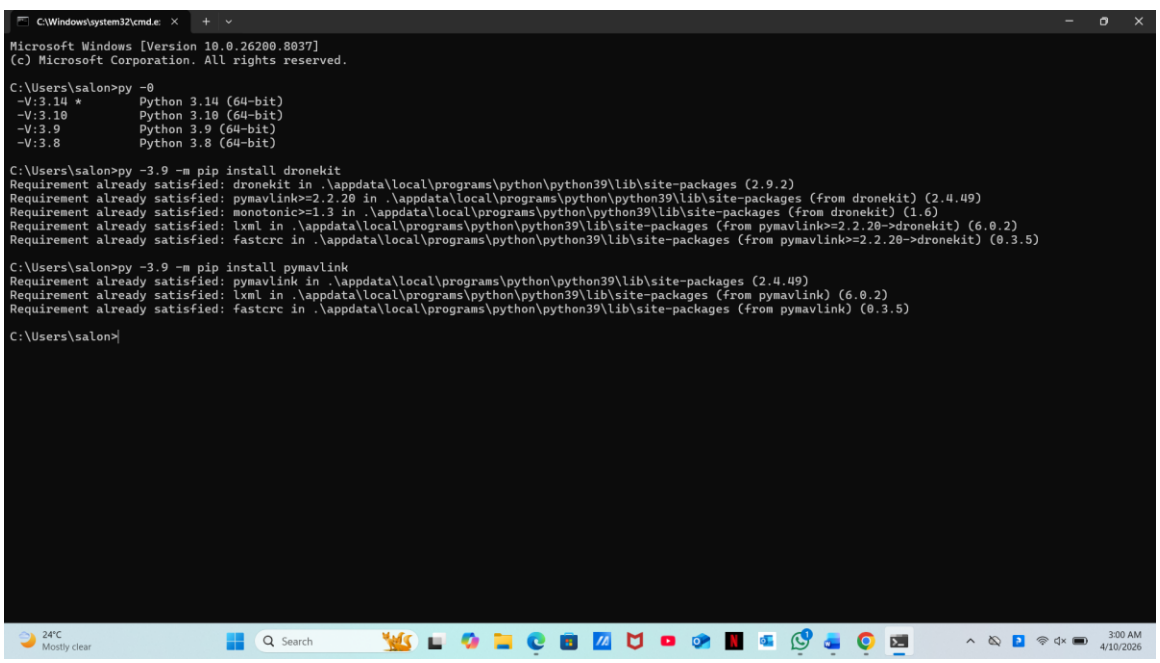
"Autopilot Firmware" refers to two distinct technologies depending on the context: **Drone/UAV flight control** or **Microsoft Windows Device Management**.

In the context of drones, Unmanned Aerial Vehicles (UAVs), and robotics, autopilot firmware is the specialized software installed on flight controller boards.

## Part 2 – Environment Setup (Simulation)

Install Python 3.9 or above.
Then install required libraries:
pip install dronekit
pip install pymavlink
Use ArduPilot SITL for simulation.

- **Windows:** Download the latest stable installer (e.g., Python 3.12) from the [Python.org Downloads page](). During installation, **ensure you check the box "Add Python to PATH.**
- **After that we go to command prompt and install the required libraries.**
- **That is dronekit and pyamavlink.**
- **And use Ardupilot SITL for simulation.**



## Part 3 – First Drone Connection Script

Python Code:

```
from dronekit import connect
print("Connecting to vehicle...")
vehicle = connect("127.0.0.1:14550", wait_ready=True)

print("Drone is connected")
print("GPS:", vehicle.gps_0)
print("Battery:", vehicle.battery)
print("Last Heartbeat:", vehicle.last_heartbeat)

vehicle.close()
print("Connection closed")
```

This script connects to a simulated drone and reads GPS, battery, and heartbeat data.

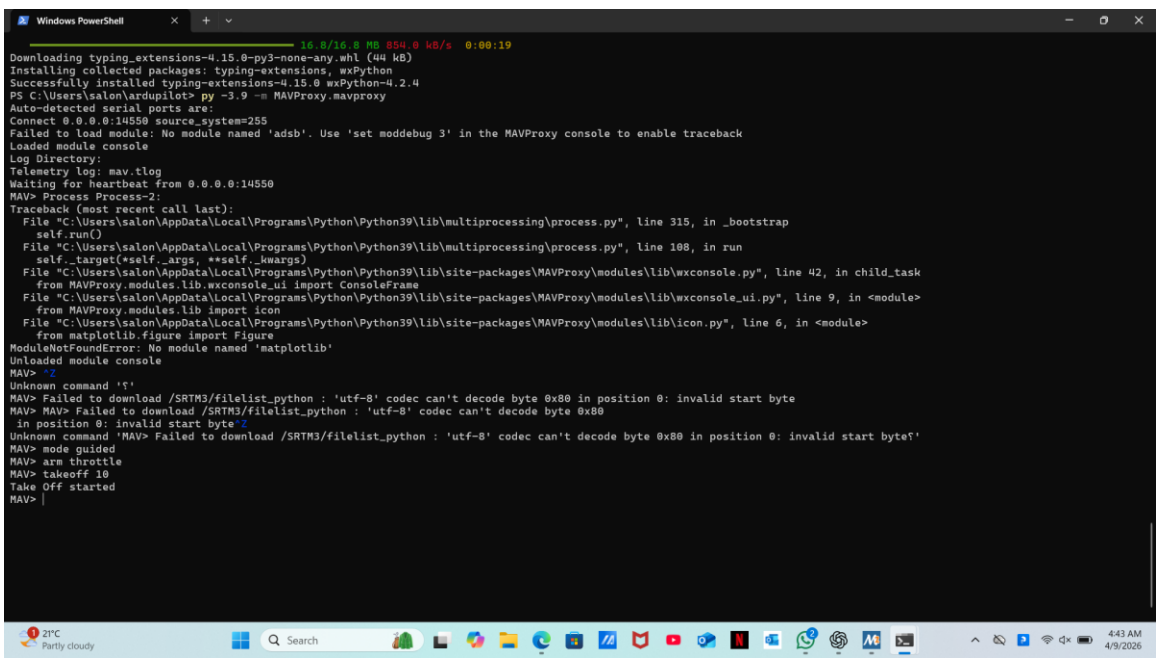
```
 Directory of C:\Users\salon

03/31/2026  07:43 PM    <DIR>          .
02/17/2026  12:21 AM    <DIR>          ..
03/31/2026  07:43 PM    <DIR>          .dronekit
02/15/2026  06:14 PM    <DIR>          .vscode
02/18/2026  01:59 AM    <DIR>          anaconda3
08/08/2025  09:40 AM    <DIR>          Contacts
02/18/2026  10:37 PM    <DIR>          Desktop
09/19/2025  08:10 PM    <DIR>          Documents
03/31/2026  07:38 PM    <DIR>          Downloads
03/31/2026  07:40 PM    <DIR>          drone_env
08/08/2025  09:40 AM    <DIR>          Favorites
02/14/2026  01:28 PM    <DIR>          iu-14D2N.tmp
08/08/2025  09:40 AM    <DIR>          Links
08/08/2025  09:40 AM    <DIR>          Music
08/08/2025  09:47 AM    <DIR>          OneDrive
11/17/2025  12:03 AM    <DIR>          Pictures
08/08/2025  09:40 AM    <DIR>          Saved Games
08/08/2025  09:58 AM    <DIR>          Searches
02/18/2026  10:05 AM    <DIR>          Videos
               0 File(s)              0 bytes
              19 Dir(s)  400,792,723,456 bytes free

(drone_env) C:\Users\salon>cd C:\Users\salon

(drone_env) C:\Users\salon>drone_env\Scripts\activate

(drone_env) C:\Users\salon>python connect_drone.py
Connecting to vehicle...
CRITICAL:autopilot:APM:Copter V3.3 (d6053245)
CRITICAL:autopilot:Frame: QUAD
CRITICAL:autopilot:Calibrating barometer
CRITICAL:autopilot:Initialising APM...
CRITICAL:autopilot:barometer calibration complete
CRITICAL:autopilot:GROUND START
Drone is connected
GPS: GPSInfo:fix=3,num_sat=10
Battery: Battery:voltage=12.587,current=0.0,level=100
Last Heartbeat: 0.7189999999991414
Connection closed
```

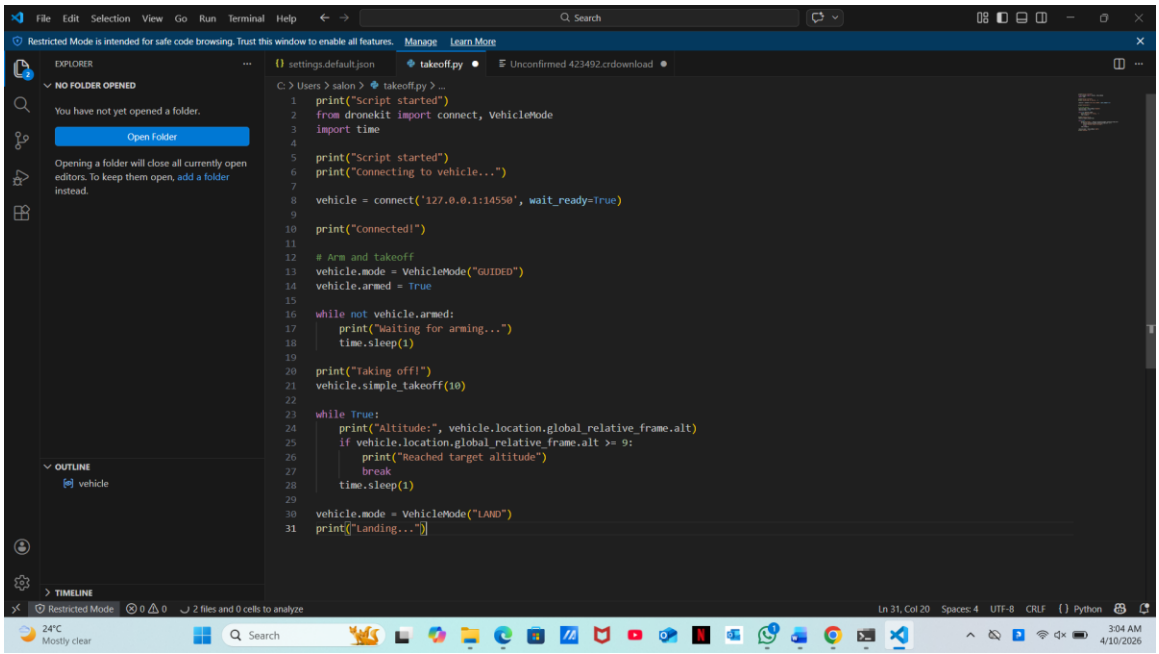
## Part 4 – Basic Arm and Takeoff Script

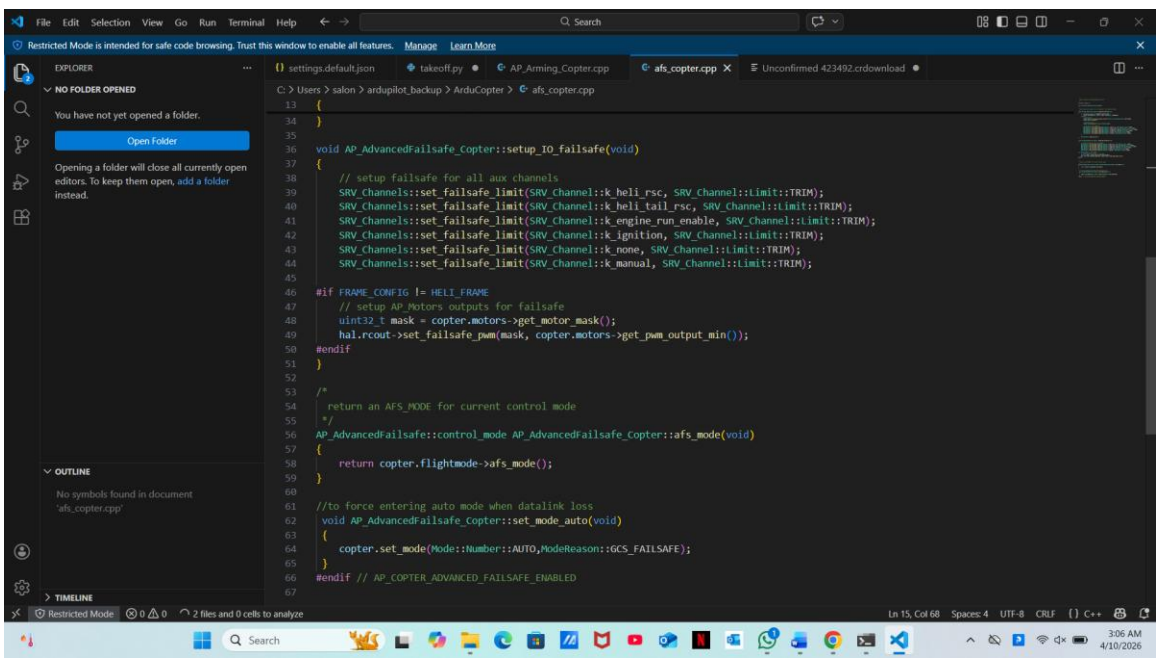
Python Code:

from dronekit import connect, VehicleMode

```
import time

vehicle = connect("127.0.0.1:14550", wait_ready=True)

def arm_and_takeoff(target_altitude):
    print("Arming motors")
    vehicle.mode = VehicleMode("GUIDED")
    vehicle.armed = True

    while not vehicle.armed:
        print("Waiting for arming...")
        time.sleep(1)

    print("Taking off!")
```






# PART 5 – MINI PROJECT

A waypoint is a specific, intermediate geographic coordinate (latitude and longitude) used in navigation to define a route, mark a location, or track movement. Often used in GPS systems, aviation, and maritime travel, waypoints act as "signposts" that guide travelers from their origin to a final destination.

Autonomous navigation is the ability of a vehicle or robot to move, plan its path, and avoid obstacles without human intervention, using onboard sensors and AI algorithms. It enables self-sufficiency in tasks ranging from robotic transport to self-driving cars, making decisions based on real-time environmental data.